

Predicting Music Genre from Audio Features: A Machine Learning Approach

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Introduction

Music genre classification is a challenging task due to the subjectivity of musical features and the inherent overlap between different genres. However, accurately predicting the genre of a song can have many useful applications in the music industry. For example, music streaming platforms can use genre predictions to improve their recommendation systems, allowing users to more easily discover new music that aligns with their tastes. Automatic genre classification can also be useful for radio stations, music television networks, and other media outlets that rely on genre labels to organize and present their content. In addition, genre predictions can help musicians and music producers better understand the characteristics of different musical styles and how they can incorporate these features into their own compositions.

In this research project, we aim to predict the genre of a song based on its musical features using a random forest tree classifier. To do this, we collected a dataset of songs from Spotify and their corresponding musical features and genre labels. We then explored the data to understand the relationships between the different musical features and genres and selected an appropriate machine learning model to train on the data.

Related Work

This research project was largely inspired by *The Personality of Music Genres* (Neuman et al.) and *Personality, Uses of Music, and Music Preference: The Influence of Openness to Experience and Extraversion* (Vella & Mills).

Both papers [2, 3] delve into the connections between personality and music, using different techniques to analyze music genres and correlate them to different measures of personality. Exploring the relationship between music and genre helped better understand the topic domain and ultimately led to the problem of quantifying our music genres. If we can find a way to quantify music genre, in our case, through classification using musical features, then we can expand on the applications of music genre vs personality.

Data Collection

For this study, we used a Kaggle dataset [3] sourced from Spotify's public API. The dataset contained 6917 rows, each representing a song in one of seven selected genres: *pop*, *rock*, *latin*, *hip-hop*, *edm*, *rap*, and *r&b*. The dataset included various numeric features calculated by Spotify, such as the song's *popularity*, *danceability*, *energy*, and *loudness*, as well as metadata such as *the genre*, *title*, *album cover link*, *artist*, and *ID*.

We selected this dataset because it provided a diverse range of music genres and many audio features that could be used to classify the genres. A detailed description of the dataset and its features can be found in the appendix.

Data Preparation

Before analyzing our dataset, we performed several preprocessing steps to clean and normalize the data.

First, we normalized the numerical feature values by setting the mean to 0 and the standard deviation to 1. This helps to scale the data and make it more suitable for analysis.

Next, we cleaned and formatted the dates in the dataset to make them more readable and consistent. The original dataset included two columns for the date, one for the time period (day/week/month/year) and the other with a string representation of the time (e.g., '01-2018'). We subsetting each string to only include the year and replaced the two columns with a single variable called *release_year*.

We also removed unnecessary features from the dataset, including the *song ID* and *album cover links*.

Finally, we split the dataset into three separate datasets for further analysis: a development set (20% of the data), a cross-validation set (70% of the data), and a final test set (10% of the data). This allowed us to evaluate the performance of different algorithms on different subsets of the data and ensure that the results were robust and reliable.

Discussion of Class Value

Our dataset includes a *genre* variable, which we will use for training our genre classifications. The class value comes from the Spotify API, where its value is derived from a combination of both manual and automated factors.

The manual curation process involves human experts who review and classify songs into genres based on their musical characteristics and style. In addition to manual curation, Spotify also uses automated algorithms to classify songs into genres.

As we move forward, we keep in mind the limitations of using a class value that is partially created automatically. In the future, we hope to use a more accurate source of measuring genres.

Baseline Performance and Error Analysis

i. Baseline Performance

We begin predicting our songs' genres using a Random Forest Tree Classifier, inputting all our variables as predictors. A model evaluation matrix and confusion matrix are shown below:

Metric	Value
Accuracy	0.5883
Kappa	0.5143

Act \ Pred	edm	hiphop	latin	pop	r&b	rap	rock
edm	147	0	5	31	0	2	4
hiphop	1	93	6	14	5	70	2
latin	9	10	115	39	9	6	14
pop	32	2	20	142	11	22	15
r&b	2	13	24	37	20	11	16
rap	1	69	5	22	1	97	3
rock	9	2	4	16	3	2	199

ii. Error Analysis

Identification of Problematic Features:

To better understand the performance of the classifier, we conducted an error analysis to identify problematic features and classes. We observed that the classifier had a particularly difficult time distinguishing between *hip-hop* and *rap*, as well as classifying *r&b* songs.

To further investigate these issues, we conducted a horizontal and vertical analysis of the features for these three classes. The horizontal analysis showed that the features with the highest absolute difference when a hip hop song was misclassified as a rap song were year and popularity. Similarly, when a rap song was misclassified as a hip hop song, *year* and *valence* had the highest difference. *Year* also had a high difference with *r&b* in general.

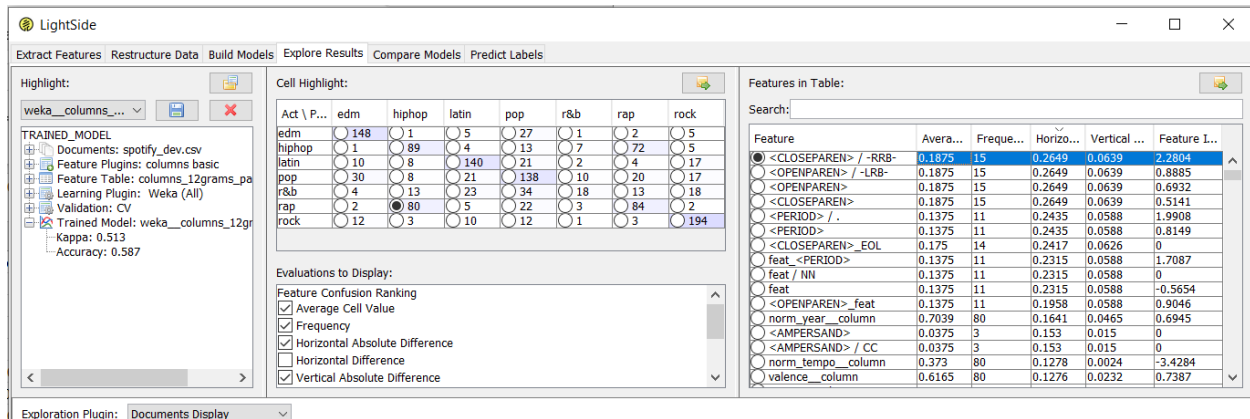
The vertical analysis showed that a mix of numerical features affected the predictions for these classes. *Valence*, *song duration*, and *danceability* had low vertical differences, while other features such as *year* and *tempo* had higher differences.

These results suggest that some of the features may be less useful for distinguishing between these classes, and further investigation may be necessary to improve the performance of the classifier on these classes.

Horizontal Difference Examples:

The screenshot displays the LightSide software interface, which is used for analyzing classifier performance. The interface is divided into several panels:

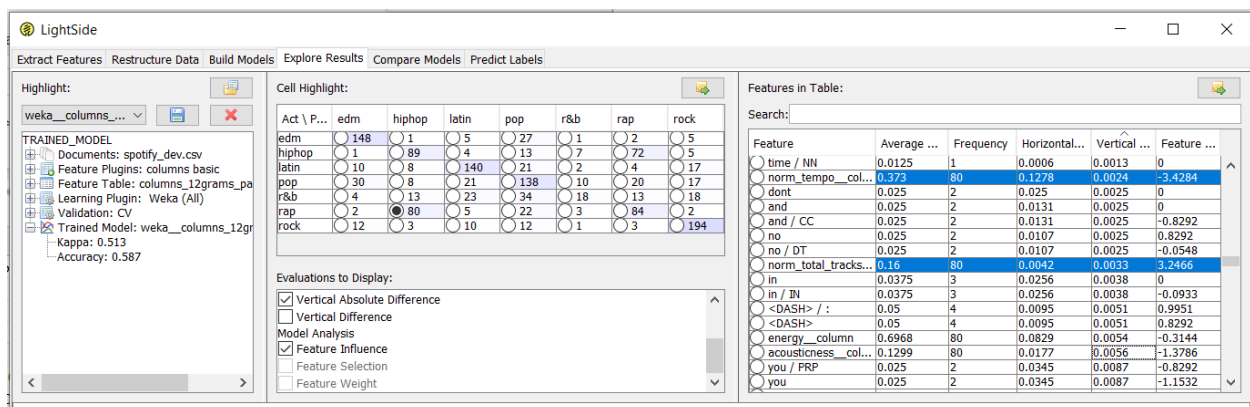
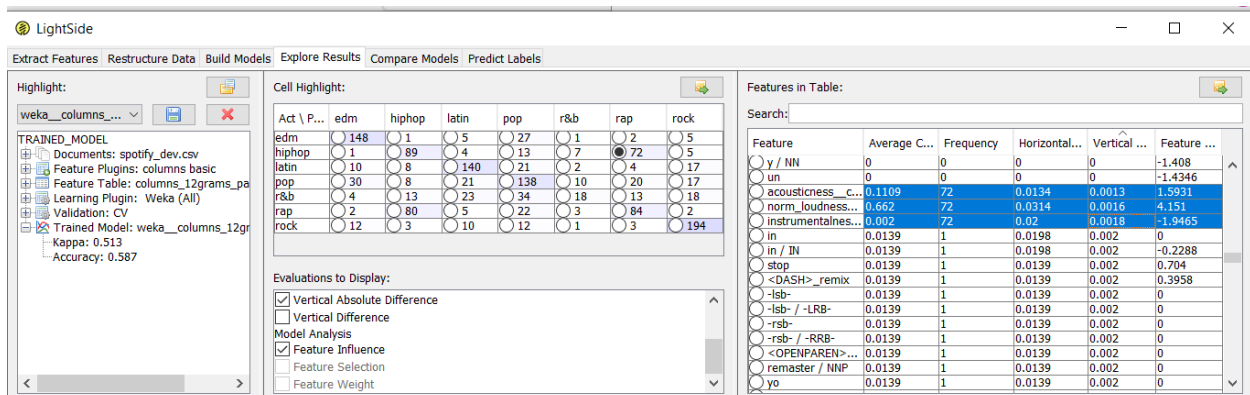
- Highlight:** Shows the selected dataset, `weka_columns_...`.
- Cell Highlight:** Displays a confusion matrix for the classifier. The matrix shows the relationship between actual classes (edm, hiphop, latin, pop, r&b, rap, rock) and predicted classes. The diagonal elements represent correct classifications, while off-diagonal elements represent misclassifications. For example, the 'rap' row shows a high number of misclassifications into 'hiphop' (69) and 'edm' (1).
- Evaluations to Display:** A list of evaluation metrics that can be displayed, including:
 - Feature Confusion Ranking
 - Average Cell Value
 - Frequency
 - Horizontal Absolute Difference
 - Horizontal Difference
 - Vertical Absolute Difference
- Features in Table:** A table showing the results of the feature analysis. The table includes columns for Feature, Average C..., Frequency, Horizontal ..., Vertical Ab..., and Feature Inf... The table lists various features and their corresponding values, highlighting the features that have the highest absolute differences in the horizontal and vertical analyses.



We find that parentheses and “feat” have high horizontal differences for both rap and hip-hop errors. This could be explained by hip-hop and rap songs commonly featuring other artists. Usually, the features are in the form, “[Song Name] (feat. [Artists])”, so the parentheses and “feat” are caught in our horizontal analysis.

The word “you” also commonly has a high horizontal difference.

Vertical Analysis:



In terms of vertical differences, our numerical features in general show highly weighted, low vertical differences.

Note that in this case, we don't identify text features as problematic because none of them have frequencies high enough to have a major effect on our predictions. Most text features occur at most 3 times.

In real life, hip-hop songs commonly contain rap verses, which may explain the ambiguity between the two genres because features such as speechiness and danceability are homogenous between the two genres. Our song titles are also similar, with many song titles containing the phrase "(feat. [artist name])".

R&B is also hard to classify for us. In real life, it's a very broad genre that contains features from other genres. We notice from vertical analysis that our numeric features in general are very low – R&B has trouble having a solid identity (with our limited features, at least) that distinguishes it from other genres.

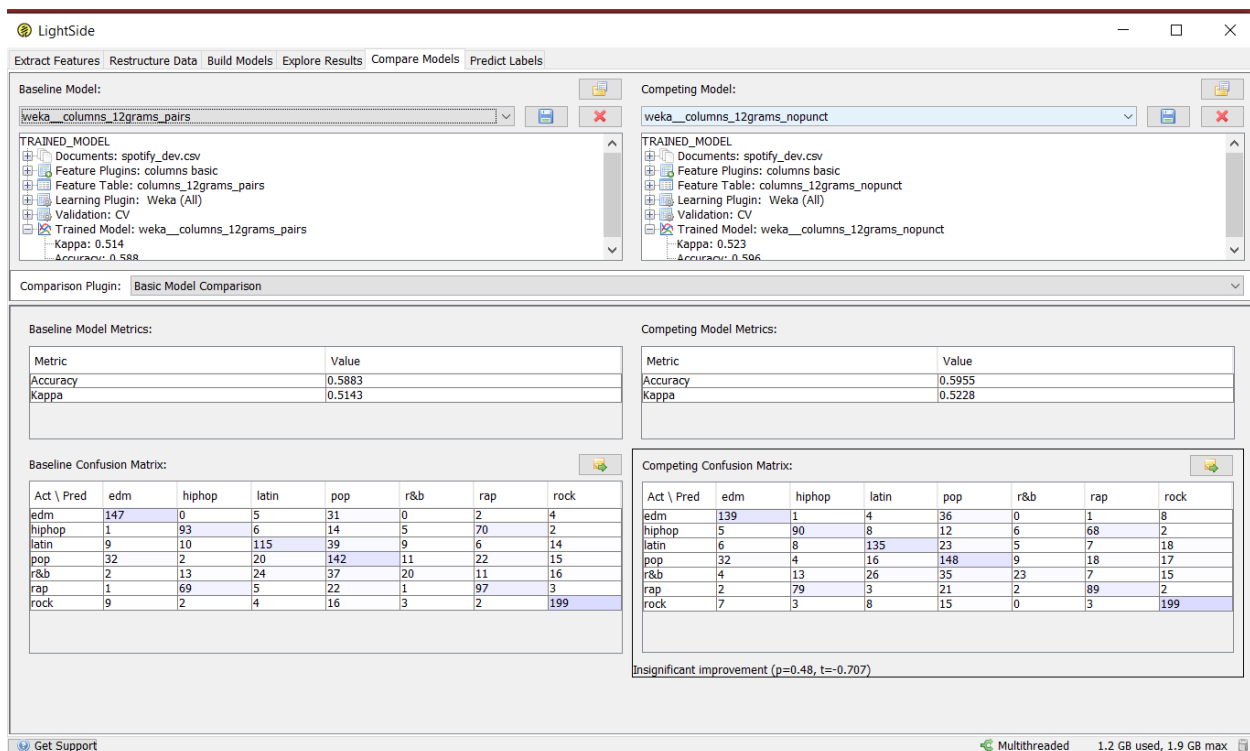
Potential Strategies for Improving Performance:

To improve the performance of the classifier on certain classes, we propose several potential strategies for further investigation.

First, we plan to modify the song names by removing text inside parentheses. This is intended to eliminate the influence of artist names on the classification process, as it is common for song names to include artist features in parentheses. By focusing on the song names themselves, we hope to gain a better understanding of the characteristics that distinguish different genres.

In addition, we will consider removing punctuation and potentially stop words from the song names. Some words, such as "if", "you", and "me", are very general and could be used in any genre's song names, while other words like "Edit", "Remastered", and "Gang" may provide more specific indications of genre. Removing stop words may help to emphasize the importance of these genre-specific words and improve the performance of the classifier.

Evaluation of Ideas:



We find that our proposed fixes do not have a significant improvement, where our p-value of 0.48 and our t-statistic is 0.707. It's possible that our features don't give us enough information to distinguish between hip-hop and rap. In the future, we hope to incorporate lyric data into our dataset. We predict that with lyric data, distinguishing between the two will be much easier. It can possibly help us with distinguishing r&b as well.

Optimization and Tuning

To improve the performance of our Random Forest classifier, we explored the effect of adjusting the parameter numFeatures, which controls the number of features that are randomly selected from the dataset for each decision tree in the forest. We tested four different values for numFeatures: 0 (corresponding to the formula $\text{int}(\log_2(\#\text{predictors}) + 1)$ in Weka, which is approximately 4 in our case), 2, 5, and 10.

The baseline performance of the classifier using the default value of 0 for numFeatures was a Kappa statistic of 0.51.

We split our validation dataset into a training set and a testing set and created 5 folds for each of the sets. We used manual cross-validation to identify the optimal numFeatures setting for each fold in the training set.

Running our optimal settings from our training data on our test set, we get the following results:

Fold	Train Set: numFeatures = 4	Train Set: numFeatures = 2	Train Set: numFeatures = 5	Train Set: numFeatures = 10	Optimal Setting	Test Set Performance
1	0.41	0.42	0.44	0.45	10	0.44
2	0.55	0.52	0.55	0.52	5	0.44
3	0.44	0.40	0.45	0.43	5	0.46
4	0.46	0.42	0.43	0.40	4	0.44
5	0.48	0.49	0.48	0.49	2	0.49

We then applied the optimal numFeatures settings from the training data to the corresponding folds in the testing set and calculated the performance of the classifier. Overall, the performance of the classifier using the optimal numFeatures settings was not statistically significantly different from the baseline performance, as determined by a two-tailed paired-T-test with a p-value of 0.396.

These results suggest that adjusting the numFeatures parameter may not have a significant impact on the performance of the Random Forest classifier in our specific dataset. Further investigation may be necessary to identify other factors that could contribute to the performance of the classifier.

Result

Performance

Since our tuned models and error analysis did not show any significant improvement from the baseline model, we continue with our default parameters on our testing data.

Using a random forest classifier model, with 4 randomly selected features, we get a final Kappa statistic of 0.5247 and an accuracy of 59.60%.

Overall, we find that our random forest model performs significantly well on determining songs in the edm, latin, pop, and rock genres. However, we struggle heavily to distinguish between hip-hop, r&b, and the rap genre. Below is a table demonstrating the performance on each genre, using percentage correct:

edm	hiphop	latin	pop	r&b	rap	rock
0.783	0.514	0.588	0.686	0.164	0.411	0.864

Weighted average: 0.596

Discussion and Limitations

We found through error analysis that our model is largely flawed in distinguishing between certain genres of music, namely hip-hop and rap. We believe that the partially automated classification of our class value may have played a role in making these two genres so hard to distinguish.

In the future, we hope to add more features, such as song lyrics, to our dataset in hopes to help give more insights into the differences between certain music genres.

By analyzing musical features of songs of different genres, we gained insights into the specific features of different genres. For example, we found that words commonly in song titles such as "Edit," "Remastered," and "Gang" provides specific indications of their genres. Overall, we learned niche details about a song's title and musical features to help distinguish it from songs in other genres. We also made relatively successful progress in training a classification algorithm to distinguish between genres.

References

- [1] Neuman, Y., Perlovsky, L., Cohen, Y., & Livshits, D. (2016). The personality of music genres. *Psychology of Music*, 44(5), 1044–1057. <https://doi.org/10.1177/0305735615608526>
- [2] Vella, E. J., & Mills, G. (2017). Personality, uses of music, and music preference: The influence of openness to experience and extraversion. *Psychology of Music*, 45(3), 338–354. <https://doi.org/10.1177/0305735616658957>
- [3] Noah (2021, December). Spotify Genre Audio features. Retrieved December 16, 2022 from <https://www.kaggle.com/datasets/naoh1092/spotify-genre-audio-features>.
- [4] Web API Reference | Spotify for Developers. (n.d.). Web API Reference | Spotify for Developers. Retrieved December 16, 2022, from <https://developer.spotify.com/documentation/web-api/reference/>

Appendix

[1] A description of each of the heuristics is shown below, sourced from Spotify's Web API Guide [4]:

- **Danceability:** Describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity.
- **Valence:** Describes the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry).

- **Energy:** Represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale.
- **Tempo:** The overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece, and derives directly from the average beat duration.
- **Loudness:** The overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing relative loudness of tracks.
- **Speechiness:** This detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value.
- **Instrumentalness:** Predicts whether a track contains no vocals. “Ooh” and “aah” sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly “vocal”.
- **Liveness:** Detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live.
- **Acousticness:** A confidence measure from 0.0 to 1.0 of whether the track is acoustic.
- **Key:** The estimated overall key of the track. Integers map to pitches using standard [Pitch Class notation](#) . E.g. 0 = C, 1 = C#/Db, 2 = D, and so on.
- **Mode:** Indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0.
- **Duration:** The duration of the track in milliseconds.
- **Time Signature:** An estimated overall time signature of a track. The time signature (meter) is a notational convention to specify how many beats are in each bar (or measure).

[2] A sample of our dataset is shown below:

Genre	Title	Artist	explicit	popularity	release_year	total_tracks	danceability	energy	key	loudness	mode	speechiness	acousticness	instrumentalness	liveness	valence	tempo	time_signature	duration_min	norm_popularity	norm_year	norm_total_tracks	norm_loudness	norm_tempo	norm_duration_min	
pop	Pretty Girl Rock			Keri Hilson	FALSE	69	2010	12	0.666	0.87	11	-5.004	0	0.246	0.203	0	0.0851	0.9	160.0144	4.065333333	0.69	0.807017544	0.120879121	0.752582592	0.70566724	0.299091724
	If I Aint Got You			Alicia Keys	FALSE	80	2003	15	0.609	0.444	7	-9.129	1	0.106	0.603	0.00000686	0.104	0.166	118.4023	3.811766667	0.8	0.684210526	0.153846154	0.541628311	0.433446072	0.261398041
	Bad Life (feat. Kali Uchis)			Omar Apollo	FALSE	67	2021	1	0.565	0.25	11	-8.927	0	0.0306	0.743	0.0000726	0.109	0.163	109.0484	3.2969	0.67	1	0	0.551958679	0.372253224	0.184861083

rap	The Real Slim Shady	Eminem	TRUE	85	2000	18	0.949	0.661	5
	-4.244	0	0.0572	0.0302	0	0.0454	0.76	104.5044	4.736666667
	0.631578947	0.186813187	0.79144932	0.342526871	0.398888069				0.85
hiphop	Rappers Delight	The Sugarhill Gang	FALSE	44	1999	19	0.9	0.845	11
	-7.306	0	0.257	0.465	0	0.0724	0.914	111.1874	7.122666667
	0.614035088	0.197802198	0.634857318	0.386246328	0.753576366				0.44